Project Report Format of Science Exhibition to be filled by Student

Brief about the work undertaken (in about 600 - 1,200 words) in the following format.

Student Names:	**Gautam Kumar & Abhinav Kumar** (Class: Class IX, Category II (Classes 9–11))
School & Mentor:	Udaan International School, Begusarai, Bihar | Mentor Teacher: **Soni Kumari (PGT)**
Academic Year & Project:	2026–27 | **KRISHI-DRISHTI**
Registered CBSE Sub-Theme:	**Artificial Intelligence (AI) for Better Life** (Team ID: **T26706431**)

I. Title: *(filled in by student, max words 40)*

KRISHI-DRISHTI: A Portable Embedded Camera and Edge AI Prototype for Early Foliar Disease Screening in Tomato Crops for Smallholder Farmers

II. Nature of work: *Tick one of the 03 categories-*

[ ] *Application of basic principles of science and technology*

[X] *Novel solution/ indigenous design to a problem/ challenge* **<-- Selected Project Category**

[ ] *Investigation based study*

III. Rationale: *A few sentences explaining the need for the work.*

During the nationwide tomato crisis when retail prices crossed Rs 200/kg across India, widespread crop losses were reported across major tomato-growing belts due to extreme weather and destructive foliar diseases like Early Blight and Late Blight. In smallholder farming districts like Begusarai, tomato growers routinely suffer 30% to 50% harvest losses from these fungal pathogens. Marginal farmers lack rapid access to certified plant pathology laboratories, and cellular internet signals in open agricultural fields are often too weak for cloud-based diagnostic apps. By the time symptoms become obvious, infection has already spread. Out of panic, farmers spray expensive chemical fungicides blindly, escalating debt and damaging soil ecology without curing the crop. We designed KRISHI-DRISHTI to address this critical gap: an affordable, handheld camera wand that captures sharp leaf photos and gives instant foliar screening right in the field, operating 100% offline without cellular data or recurring subscriptions.

IV. Scientific Principle(s)/ Concepts: *Specify the Science concept used/ explored*

1. Plant Pathology & Blight Symptoms (Biology): Foliar pathogens leave distinct visual patterns. Early Blight creates dark circular spots with concentric target-board rings and yellow edges. Late Blight spreads quickly, causing water-soaked dark patches that rot leaves in humid air (>85% RH). Catching these patterns early allows spot-treating infected areas before entire fields are lost.

2. Optics & CMOS Camera Sensing (Physics): To see tiny fungal spots clearly, we used an OV5640 5-megapixel sensor with an adjustable macro lens. Light bouncing off leaves hits light-sensitive pixels through a Bayer filter, converting photons into electrical signals to capture sharp 1600×1200 JPEG photos.

3. Smart Memory & Direct Storage (Electronics & Computing): High-resolution photos take substantial memory. The ESP32 uses Direct Memory Access (DMA) to stream images into a 4 MB PSRAM chip, saving them over SPI to a microSD card while driving an SSD1306 OLED screen over I2C to show live status updates.

4. Edge AI & On-Device Screening (Applied AI): Instead of uploading photos to cloud servers, screening runs locally on edge hardware. A lightweight MobileNet neural network scans leaf patterns, spot shapes, and color edges in milliseconds, working entirely offline without internet.

V. Materials Used:

• List of materials that are used for developing the exhibit:

- ESP32-CAM Development Board (dual-core 240 MHz processor, 520 KB RAM, 4 MB external PSRAM).
- OV5640 5-Megapixel CMOS Camera Module with adjustable manual macro lens for sharp close-ups.
- 8 GB MicroSD Card (Class 10, FAT32) for saving leaf images directly in the field.
- SSD1306 0.96-inch 128×64 I2C OLED Display Module for real-time status and capture alerts.
- Portable 5V USB Power Bank (10,000 mAh rechargeable Li-ion battery) with micro-USB cable.
- Handheld wand handle and protective enclosure built to hold the camera steady against leaves.
- Botanical tomato leaf specimens: Healthy foliage and naturally infected leaves showing blight spots.

• Any measuring instruments/ equipment/ tools required for operating the exhibit:

- Digital Multimeter (used for checking continuity and verifying steady 3.3V power rails).
- FTDI USB-to-UART Serial programmer module (for flashing firmware and serial debugging).
- Android Smartphone (used as the offline edge host to run the local neural network).
- Computer workstation running Arduino IDE 2.x and Python 3.11 analytical tools.

VI. Procedure/ Description:

• Sequentially numbered steps that cover the procedure from beginning to end (detailed enough for replication):

Step 1: Circuit Assembly & Wiring — We connected the OV5640 camera ribbon to the ESP32-CAM and wired the OLED display via I2C (GPIO 14/15). We used a multimeter to verify safe, steady 3.3V power levels across all pins.

Step 2: Camera Code & Memory Setup — We programmed the firmware in Arduino IDE, setting the camera to 1600×1200 UXGA resolution. We routed image frames into external PSRAM so the microcontroller never runs out of memory.

Step 3: MicroSD Storage Setup — We set up the 8 GB microSD card with automatic file naming (/photo_0001.jpg) and write verification to ensure each picture saves completely.

Step 4: Farmer Display Interface — We programmed the OLED screen to show clear, simple alerts: camera ready, SD card ready, and photo saved confirmation.

Step 5: Offline Edge Link — We set up a direct local SoftAP Wi-Fi connection between the wand and an Android phone to transfer photos instantly without needing cellular towers or mobile data.

Step 6: Macro Lens Calibration — We calibrated the manual focus lens by photographing leaves from 5 cm to 25 cm away, finding that 10 to 12 cm gives the sharpest view of fungal lesion rings.

• *Sub-divided sections of the procedure (variables, hypothesis, safety precautions, experimentation):*

[Variables and Parameters]

• Independent Variables: Camera distance from leaf (5 cm to 25 cm), lighting (direct sunlight vs. shaded daylight), and leaf condition (healthy green vs. blight-spotted leaves).
• Dependent Variables: MicroSD write duration (ms), JPEG file size (KB), lesion edge sharpness, and edge AI screening confidence (%).
• Controlled Variables: Sensor resolution (1600×1200 UXGA), JPEG quality setting (10), supply voltage (regulated 3.3V), and SD clock speed.

[Scientific Hypothesis]

If a 5-megapixel camera sensor with manual macro focus is paired with a PSRAM-backed ESP32 at 10 to 12 cm from a leaf, it will capture uncorrupted 1600×1200 photos in under a second and store them locally, providing the clear visual detail needed for offline edge AI screening without cellular internet.

[Safety Precautions]

• Electrical Safety: Checked power rails with a multimeter before connecting to ensure 3.3V sensitive camera pins never received excess voltage.
• Battery Safety: Used a protected power bank in a ventilated case to prevent overheating during outdoor testing.
• Biological Hygiene: Wore disposable gloves when handling diseased leaves and sealed samples in zip-lock bags after testing to prevent spores spreading to garden plants.

[Experimentation & Current Implementation Status (Technical Honesty Statement)]

• Hardware-Verified Features: The ESP32-CAM initializes the OV5640 sensor (PID 0x5640), buffers frames in 4 MB PSRAM, saves uncorrupted 1600×1200 images to the SD card in under a second, and shows live OLED diagnostics on battery power.
• Ongoing Work (Scientific Honesty Note): The MobileNet model is trained and validated on benchmark datasets on a computer. Full integration into a standalone Android APK for farmer field testing is currently underway.

• *Sub-divided sections of the procedure (data analysis, results, conclusions and implications):*

[Data Analysis & Observations]

Bench testing showed the prototype captures and writes 1600×1200 frames in 850 to 1,050 ms, producing JPEGs between 180 KB and 320 KB. Optical tests confirmed that keeping the camera 10 to 12 cm away produces sharp images where concentric rings down to 1.5 mm across are clearly visible, whereas distances below 6 cm caused macro blur and direct midday sun created washed-out glare spots.

[Results, Limitations, and Practical Conclusions]

The prototype proves that an offline, handheld crop screening tool can be built from open hardware for under Rs 1,500 (unit BOM: Rs 1,480).

Practical Limitations: (a) Leaf imaging only spots visible surface symptoms and cannot detect root rot; (b) Bright outdoor sunlight creates glare spots, so leaves should be shaded by hand; (c) Edge AI provides an early alert screening, not a certified lab test.

[Agricultural Implications & Future Improvements for Small Farmers]

KRISHI-DRISHTI shows that AI can assist rural farmers without expensive equipment or continuous 5G internet. Early screening allows timely spot-treatment instead of blanket spraying, saving money and protecting soil. Future work includes models for potato and chili, Hindi/Maithili audio alerts, and a rain-resistant field case.

VII. References:

- Agrios, G. N. (2005). Plant Pathology (5th ed.). Academic Press / Elsevier.
- Indian Council of Agricultural Research (ICAR) - National Research Centre for Integrated Pest Management (NCIPM). Package of Practices for Tomato Pest and Disease Management. New Delhi.
- Food and Agriculture Organization of the United Nations (FAO). (2021). The State of Food and Agriculture: Making Agrifood Systems More Resilient. Rome.
- Howard, A. G., et al. (2017). MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications. arXiv:1704.04861.
- Central Board of Secondary Education (CBSE). (2026). Guidelines and Information Brochure: CBSE Science Exhibition 2026–27 (Circular No. Acad-69/2026). Delhi.

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